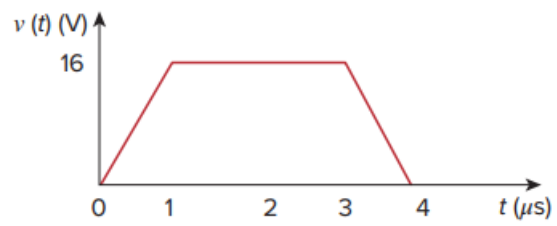


Problem

The voltage across a 5-mF capacitor is shown in Fig. 6.47. Determine the current through the capacitor.

Fig. 6.47:



Step-by-step solution

Step 1 of 6

Consider the following voltage wave as shown in Figure 1.

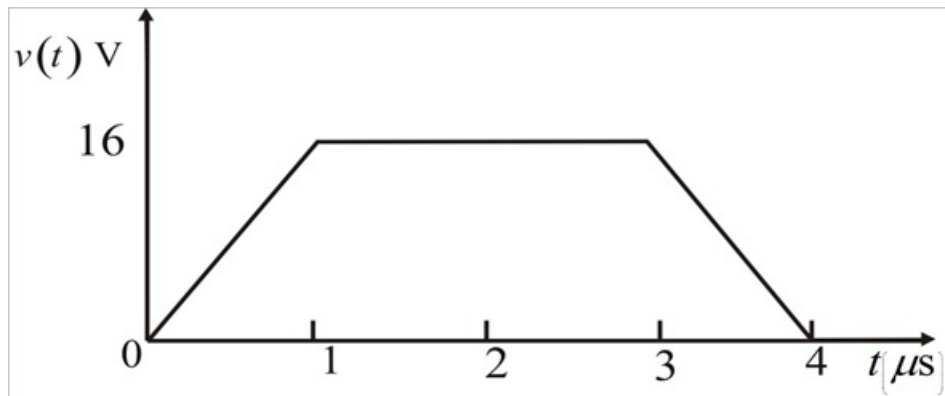


Figure 1

Step 2 of 6

Given capacitor is $C = 5 \text{ mF}$

Write down the expression for voltage $v(t)$ for the interval $0 < t < 1\mu\text{s}$.

$$v(t) = 16t$$

Write down the expression for voltage $v(t)$ for the interval $1\mu\text{s} < t < 3\mu\text{s}$.

$$v(t) = 16$$

Write down the expression for voltage $v(t)$ for the interval $3\mu\text{s} < t < 4\mu\text{s}$.

$$v(t) - 16 = \frac{16 - 0}{3 - 4}(t - 3)$$

$$v(t) - 16 = -16t + 48$$

$$v(t) = -16t + 64$$

$$\text{Therefore, } v(t) = \begin{cases} 16t ; 0 < t < 1\mu\text{s} \\ 16 ; 1\mu\text{s} < t < 3\mu\text{s} \\ -16t + 64 ; 3\mu\text{s} < t < 4\mu\text{s} \end{cases}.$$

[Comments \(1\)](#)



Anonymous

never include 10^{-6} inside the calculation>?

Step 3 of 6

Determine the current through the capacitor at different time intervals:

For the time interval of $0 < t < 1\mu\text{s}$, find the current $i(t)$.

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= 5 \times 10^{-3} \frac{d}{dt}(16t) \\ &= 5 \times 10^{-3} (16) \\ &= 80 \times 10^{-3} \text{ A} \end{aligned}$$

$$i(t) = 80 \text{ mA}$$

Therefore, the value of current $i(t)$ for $0 < t < 1\mu\text{s}$ is 80 mA.

Step 4 of 6

For the time interval of $1\mu\text{s} < t < 3\mu\text{s}$, find the current $i(t)$.

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= 5 \times 10^{-3} \frac{d}{dt}(16) \\ &= 0 \text{ A} \end{aligned}$$

Therefore, the value of current $i(t)$ for $0 < t < 1\mu\text{s}$ is 0 mA.

Step 5 of 6

For the time interval of $3\mu s < t < 4\mu s$, find the current $i(t)$.

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= 5 \times 10^{-3} \frac{d}{dt}(-16t + 64) \\ &= 5 \times 10^{-3}(-16) \\ &= -80 \text{ mA} \end{aligned}$$

Therefore, the value of current $i(t)$ for $3\mu s < t < 4\mu s$ is $\boxed{-80 \text{ mA}}$.

Step 6 of 6

Draw the current waveform as shown in Figure 2.

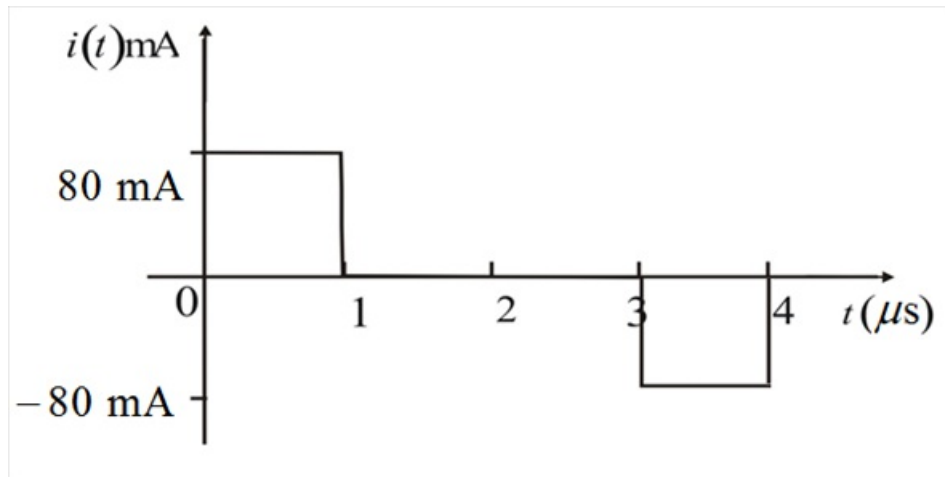


Figure 2

Hence, the required current waveform has been drawn.

[Comments \(1\)](#)



Anonymous

it is 80kA and -80kA
